

Condition Variables

Condition Variables

Learning Objectives:

- Be able to explain what a condition variable is (and is not)
- Be able to use the classic wait/signal pattern with a mutex to avoid lost wakeups
- Be able to convert a solution into a blocking one

Complimentary Reading: [OSTEP](#), Condition Variables (Chapter 30)

Motivation

- Take a look at the example `produce/consume` code from the end of class last week
 - Isn't the spinning awkward?
 - Its slow!
- Wouldn't it be nicer if the `consume` could *wait until* `produce` was ready? Wouldn't it be nicer if the `produce` code could *signal* `consume` when it is done?

Condition Variables

- Goal: a mechanism that allows multiple threads to coordinate with waiting and signaling when a condition over shared state is met.
- A Condition Variable is a queue of sleeping threads
 - The function `wait` puts the current thread into the queue
 - The function `signal` wakes up a thread from the queue
 - (There's also a function `broadcast` to wakeup all threads).
- We *associate* being on the queue with waiting for a particular condition to be true about shared state
 - Note: CondVars *are not* the actual condition!
 - CondVars just help you manage waiting and waking.
- CondVars are associated with a lock: